

EXERCISES

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3. Data Modification

1. DATA MODIFICATION

Exercise 1.1

Consider the *adidas_us_sales.rds* file, which contains sales data of Adidas in United States.

- a) Find all the sales transactions that:
 - Have a total sales amount higher than \$650,000.
 - Occurred in the cities of Miami or Los Angeles.
 - Were invoiced in the third quarter.
 - Have more than 700 units sold and a total sales amount greater than \$400,000.
 - Have a price per unit either less than \$20 or greater than \$70.
- b) Sort the sales transactions by total sales amount.
- c) Sort the sales transactions by operating profit. Which sales transaction has the lowest operating profit? And the highest?
- d) Select the columns *Retailer*, *Invoice_Date*, *Total_Sales*, and *Operating_Margin*.
- e) Incorporate four new columns into the *adidas_us_sales* data frame: *Invoice_Year* and *Invoice_Month*, representing the year and month of the invoice date, respectively; *Total_Sales_K*, indicating the total sales amount in thousands of dollars; and *Operating_Margin_Perc*, expressing the operating margin as a percentage.
- f) Find the sales transaction with the highest operating profit in the year 2021. What is the retailer and the product associated with this transaction?
- g) Find the number of sales transactions per retailer.
- h) Find the month with the highest average total sales amount.

2. CONTROL STRUCTURES

Exercise 2.1

Without using R, state the expected results given by the following chunks of code:

- a)

```
if(sqrt(9) < 2){  
  sqrt(9)  
} else {  
  0  
}
```
- b)

```
z <- 6  
if(z < 0) y <- z*3 else y <- z*5
```

```
c) x <- 15
   y <- 3
   if(x > 10)
     if(y < 2 & y != 0)
       z <- x/y

d) x <- 0
   if(x < 0) print('x is positive') else
   if(x > 0) print('x is negative') else
   print('x is 0')
```

Exercise 2.2

Create a while loop that prints all even numbers between 2 and 20.

Exercise 2.3

Write a nested loop that prints 30 numbers, grouped in 3 groups with 10 elements each, i.e. (1:10, 2:11, 3:12). The first loop should define the number of groups (3); the second loop should define the numbers to be printed, by specifying the length of each group (10). For each group, the first number should be one unit higher than the previous group. The expected result is:

```
[1] 1   [1] 2   [1] 3
[1] 2   [1] 3   [1] 4
[1] 3   [1] 4   [1] 5
[1] 4   [1] 5   [1] 6
[1] 5   [1] 6   [1] 7
[1] 6   [1] 7   [1] 8
[1] 7   [1] 8   [1] 9
[1] 8   [1] 9   [1] 10
[1] 9   [1] 10  [1] 11
[1] 10  [1] 11  [1] 12
```

Exercise 2.4

Consider the *churn.rds* file, which contains data regarding 10,000 clients of a bank, and analyse the following code snippet:

```
sal <- churn$EstimatedSalary
i <- 1
sum <- 0
while(i < length(sal)){
  if(sal[i] > mean(sal)){
    sum <- sum + sal[i]
  }
  i <- i + 1
}
sum/1e3
```

What does this code snippet do? Rewrite it without using loops, with just one line of code.

3. FUNCTIONS

Exercise 3.1

Create a function that receives a vector and returns the percentage of *NA* observations.

Exercise 3.2

Notice the following code snippet:

```
x <- 2
z <- 56.3
f <- function(x) {
  a = 34
  y = x / 4 * a * z
  y
}
```

What are the values of the variables *a*, *x* and *z* in the local environment of the function *f()*? What happened to the variable *a* after the execution of the function?

Exercise 3.3

Create a function to convert an amount between Euro (EUR) and United States Dollar (USD). This function should accept two arguments: *amount*, representing the amount to be converted, and *to*, indicating the target currency (default is EUR). Assume the conversion rate is 1 EUR = 1.08 USD.

Exercise 3.4

Create a function that takes two arguments: a numerical vector *v* and a number *x*. This function should return the percentile of vector *v* that corresponds to the value *x*. Test the function using the variable *Weekly_Sales* from the *walmart_sales.rds* file, which contains sales data of Walmart.

Exercise 3.5

Consider the following data.frame named *sales_datadf*, which contains the sales of 5 salespeople in 4 regions, similar to the one created in Exercise 4.1 from Chapter 1:

```
set.seed(10)
sales_datadf <- data.frame(
  north = sample(5000:9000, 5),
  south = sample(7000:9000, 5),
  east = sample(6500:9500, 5),
  west = sample(8500:10000, 5),
  row.names = c('salesperson1', 'salesperson2', 'salesperson3', 'salesperson4', 'salesperson5')
)
```

Create a function that identifies the regions where most of the salespeople (by default, at least 4) had sales greater than or equal to \$8,000.